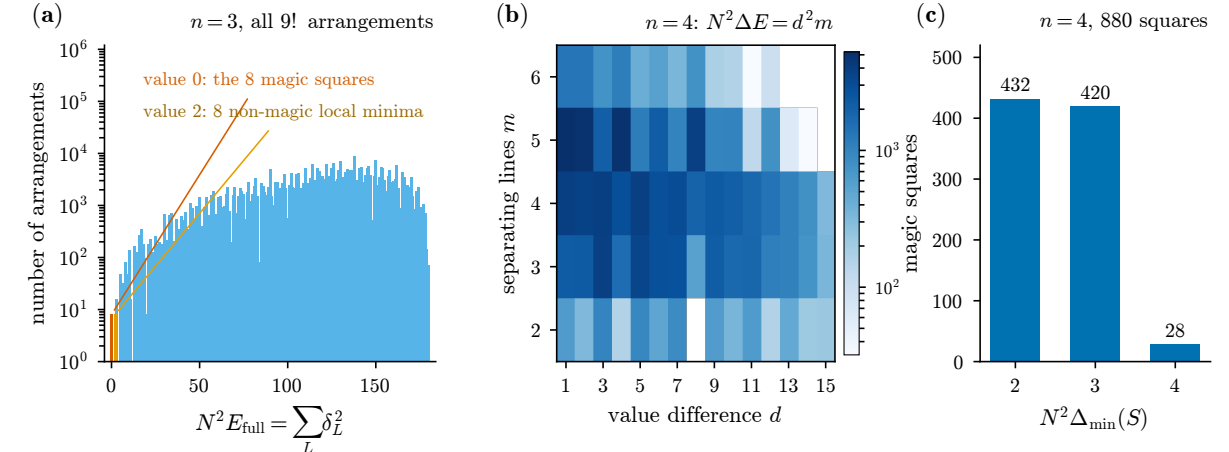
(a)
$n = 3$, all 9! arrangements
value 0: the 8 magic squares
value 2: 8 non-magic local minima
number of arrangements
10^0
10^1
10^2
10^3
10^4
10^5
10^6
0
50
100
150
$N^2 E_{\text{full}} = \sum_L \delta_L^2$
(b)
$n = 4$: $N^2 \Delta E = d^2 m$
separating lines m
2
3
4
5
6
1
3
5
7
9
11
13
15
value difference d
10^3
10^2
(c)
$n = 4$, 880 squares
magic squares
0
100
200
300
400
500
432
420
28
2
3
4
$N^2 \Delta_{\min}(S)$